

PRE-REGISTRATION

Constraint-Onset Timing and Robustness in Constrained Reinforcement Learning with the companion study *Constraint-Coverage Generalization*

Dated by the registration commit

Research group: Muhammad Umair Waseem, Hamza Nisar, Muhammad Abdullah, Abdullah,
Muhammad Talha Jamil

Supervisor: Dr. Musadaq Mansoor

Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology, Haripur, Pakistan

Registration status.

Item	Status
Document status	Adopted by the group, which is the authority for this document. Every setting carries a published source or a recorded decision; the few values marked [P] are empirical quantities that only the pilot can supply. Parts 1, 4, 5 and 6 are fixed from the go decision that follows the pilot.
Feasibility status	Registered, not yet verified. The full design of 484 run-equivalents is feasible only if the pilot's throughput measurement says so (G2 in Part 6). Until Part 8 holds that measurement and the group's machine-hour allocation, the registered design is a plan, and the cut order is the plan for the case that it does not fit.
Registration record	Complete when Part 9 holds the commit hash, its timestamp, and the public repository address. The repository is github.com/paf-iast-ai-research/when-safety-comes-late ; this document is its first commit. Until those three fields are filled, this file is the adopted draft and not the registration record.
Runs before this date	None that produce a reported result. Single-seed smoke tests of the pipeline are permitted and are not analysed.
Amendments	Any change after the go decision is recorded in Part 9 with its commit and reason. A result affected by a change made after the data were seen is reported as exploratory.

Contents

- How to Read This Document
- Part 1. Questions and Hypotheses
- Part 2. Definitions and Equations
- Part 3. Design
- Part 4. Comparison Rule
- Part 5. Analysis
- Part 6. Go or No-Go Rule and Cut Order
- Part 7. What Is New in Each Study
- Part 8. Empirical Values and Recorded Decisions
- Part 9. Registration Record and Amendment Log
- Part 10. Roles and Review
- References
- Appendix A. Settings Recorded at the Registration Commit
- Appendix B. Results Ledger Schema
- Appendix C. Targeted Search Protocol for Study B

Parts 1 to 6 follow the six required contents of the group's pre-registration note, in that order, for both studies. Parts 7 to 10 and the appendices are the further parts the note implies.

How to Read This Document

This document is a pre-registration in the sense of Nosek et al. (2018): it fixes what will be measured, how, and what would count as a failure, before any result exists. It registers two studies that share one pipeline and one analysis discipline. Study A, When Safety Comes Late, asks whether a safety constraint added late in training gives safety as robust as a constraint present from the start. Study B, Constraint-Coverage Generalization, asks whether the number and spacing of cost budgets seen during training determine how well a budget-conditioned agent generalizes to a budget it has not seen. Both are run; both have hypotheses that can fail; neither is used to support the other's claims.

Every setting has one of three marks. A setting followed by a source in parentheses is taken from that source. A setting followed by (decision) is a choice the group has made, with its reason given where it appears; it can be changed only by an amendment recorded in Part 9. A setting followed by [P] is an empirical quantity that no source and no decision can supply and only the pilot can measure; the measured value is entered in Part 8 at the go decision.

Three words are used in fixed senses. A run is one full training of one agent with one seed. An arm is one training condition; every arm is trained with five seeds. A checkpoint is the saved state of one run at one point in training. The unit of every statistical analysis is the seed. Hypotheses, rules and decisions that the group must be able to quote are set in framed boxes; quantities that have a formula are given as numbered equations in Part 2.5 and referred to by number.

Part 1. Questions and Hypotheses

1.1 Study A: the question

Question A. Does a later constraint onset cause a larger robustness gap, when the policies compared are matched on final in-distribution cost?

In the form *does X cause Y when Z*: X is the onset fraction N, the point in training at which the cost constraint is introduced; Y is the robustness gap, the rise in episodic cost when the environment is changed (equation 1); Z is the condition that final in-distribution cost has already been matched, so that no difference in Y can be a difference in how much safety was achieved.

1.2 Study A: hypotheses

Each hypothesis is stated once, exactly; the observations that support and falsify it are stated in the quantities of Part 2, and the statistical form of each rule is in Part 5. The studies are powered to establish large effects and to bound small ones, not to resolve medium ones (Part 5.5), and every claim is worded to that calibration: a supported hypothesis is one whose effect was large enough to show at this sample size, and a falsified one is one whose effect is bounded below the minimum of interest. An outcome that meets neither the supporting nor the falsifying condition is reported as inconclusive at this sample size, with its interval, and is never rounded up to support. No number in this Part is a prediction of magnitude. The same conventions apply to the hypotheses of Study B in Part 1.4.

H1. Later onset, weaker safety

Statement. At matched final in-distribution cost, a later constraint onset produces a larger robustness gap, and the gap increases with the onset fraction N.

Supported if. The gap difference $\Delta(N)$ of equation (2) is positive for $N = 0.50$ with a 95 percent interval excluding zero, and Δ increases with N, under both step-matching controls.

Falsified if. The intervals of $\Delta(0.10)$, $\Delta(0.25)$ and $\Delta(0.50)$ all include zero, or their point estimates are not ordered with N.

H2. Abrupt onset, weaker safety than a ramp

Statement. At each onset fraction greater than zero, and at matched final in-distribution cost, an abrupt onset produces a larger robustness gap than a linear ramp to the same budget.

Supported if. $\text{gap}(\text{abrupt}, N) - \text{gap}(\text{ramp}, N)$ is positive with an interval excluding zero at $N = 0.50$, and non-negative at $N = 0.10$ and 0.25 .

Falsified if. The ramp arm's gap is as large as or larger than the abrupt arm's at $N = 0.50$, or the difference's interval includes zero at every N .

H3. Plasticity-related measurements and interventions explain part of the effect

Statement. The robustness gap is associated with plasticity lost during the unconstrained phase, and interventions that restore plasticity at onset narrow the gap while additional constrained training does not. H3 tests whether plasticity-related measurements and interventions explain part of the gap; it does not claim that plasticity is the only mechanism, and it is stated as association plus intervention, not as proof of mediation.

Supported if. (a) Across late arms and seeds, the dormant-neuron fraction correlates positively and the effective rank negatively with the gap, each with an interval excluding zero. (b) Reset and injection each reduce the gap of the $N = 0.50$ arm with an interval excluding zero, and additional constrained training does not. (c) Manipulation check, at the first logging point after onset (200,000 steps): in the reset and injection arms the dormant fraction of the trainable units is lower, and the effective rank of the trainable layers' features higher, than in the untreated late arm at the same point, with intervals excluding zero. The trainable layers are the reinitialised last two layers for reset, the new head for injection, and the last two layers for the untreated arm. The check is made after 200,000 steps rather than at the moment of onset because injection changes nothing at that moment by construction. An intervention that narrows the gap without passing the check is reported as acting through something other than plasticity.

Falsified if. (a) No monotone association between the metrics at onset and the gap. Or (b) additional constrained training reduces the gap as much as reset; or reset reduces it and injection does not; or none of the three reduces it. Or (c) the manipulation check fails for both interventions.

What H3 can establish, and what it cannot: the design separates a representational explanation from a data explanation, and the manipulation check ties the interventions to the metrics they are meant to change. It cannot separate plasticity from every other change a reset or an injection makes to optimisation, and the paper says so. The word mediation is used in the paper only if all three parts hold, and then as "consistent with mediation by plasticity", not as proof.

H4. Part of the abrupt-onset effect is the controller

Statement. Part of the abrupt-onset gap is produced by the overshoot of the Lagrange multiplier: suppressing the overshoot reduces that gap but does not remove it.

Supported if. With the warm-started multiplier, the primary controller variant, the $N = 0.50$ abrupt arm's gap falls with an interval excluding zero and remains above the $N = 0$ arm's gap with an interval excluding zero. The rate-limited multiplier is a secondary check and is reported alongside.

Falsified if. With the warm-started multiplier, the difference from the $N = 0$ arm no longer excludes zero (overshoot explains all of the gap), or the gap does not fall (overshoot explains none of it).

H0. The null hypothesis of Study A

Statement. Once final in-distribution cost is matched, neither the onset fraction nor the onset shape changes the robustness gap.

Supported if. H1 and H2 are both falsified as above. H3 and H4 are then not evaluated: there is no gap difference to mediate and no abrupt-onset gap to attribute; they are reported as not tested, not as falsified.

Falsified if. H1 or H2 is supported.

1.3 Study B: the question

Question B. For a budget-conditioned PPO-Lagrangian agent trained for a fixed number of steps, does the number and spacing of cost-budget levels seen during training determine its zero-shot constraint satisfaction on an unseen budget, and the training it needs to adapt to that budget?

X is the training coverage, the set of budget levels an arm sees; Y is the constraint-satisfaction rate on an unseen budget (equation 10) and the adaptation steps needed to reach a target rate; Z is the condition that every arm trains for the same total number of steps, so that no difference in Y can be a difference in experience.

1.4 Study B: hypotheses**G1. Coverage helps**

Statement. At matched total steps, zero-shot constraint satisfaction on unseen budgets increases with the number of training levels spanning the same range: sparse (2 levels) < moderate (3) < dense (5).

Supported if. The mean satisfaction rates over the four unseen budgets are ordered sparse < moderate < dense, and dense minus sparse exceeds 5 percentage points with an interval excluding zero.

Falsified if. The three multi-level arms' means lie within 5 percentage points of one another with overlapping intervals, or are not ordered as stated.

G2. Coverage beats proximity

Statement. On each unseen budget, the dense multi-level agent satisfies the constraint more often than the single-level agent trained at the nearest budget, not only than the farthest.

Supported if. On at least three of the four unseen budgets, dense minus nearest-single exceeds 5 percentage points with an interval excluding zero.

Falsified if. The nearest single-level agent is within 5 percentage points of the dense agent, or above it, on at least two of the four unseen budgets.

G3. Stricter is harder

Statement. Extrapolation to a stricter unseen budget degrades satisfaction more than extrapolation to a looser one at the same absolute distance of 5 cost units from the training range (budget 5 below 10; budget 45 above 40). The two distances are equal in cost units and unequal in proportion, half of the lowest training budget against an eighth of the highest, and the interpretation of G3 must say so.

Supported if. For every arm, the satisfaction drop at the stricter budget (5) exceeds the drop at the looser budget (45) by more than 5 percentage points with an interval excluding zero.

Falsified if. The two drops are within 5 percentage points of each other for the majority of arms.

G4. Coverage speeds adaptation

Statement. Few-shot adaptation steps to reach the satisfaction target on an unseen budget decrease with training coverage.

Supported if. Median adaptation steps are ordered dense < moderate < sparse, and dense is at least one horizon below sparse on the majority of unseen budgets.

Falsified if. Adaptation steps of the sparse, moderate and dense arms are within one horizon of one another on the majority of unseen budgets.

G5. A few discrete levels approach the continuous case

Statement. The dense discrete arm (five levels) reaches, on unseen budgets, a zero-shot satisfaction rate within 5 percentage points of the continuous arm that samples the whole range, at the same total steps.

Supported if. Continuous minus Dense on the mean satisfaction over the four unseen budgets has a point estimate of at most 5 percentage points and an interval whose upper limit is at most 10.

Falsified if. The lower limit of the interval for Continuous minus Dense exceeds 5 percentage points: discrete coverage, however dense, does not substitute for the range.

1.5 Minimum effects of interest

Study A: 5 cost units of robustness-gap difference, one fifth of the budget of 25 (decision). A policy whose cost rises by less than a fifth of its budget under a change of environment would still pass the budget test in most deployments, so a smaller difference, even if real, would not change practice.

Study B: 5 percentage points of constraint-satisfaction rate (decision). At 100 evaluation episodes the binomial standard error of a single seed's rate is at most 5 percentage points, so a smaller difference lies within the evaluation noise of one seed and would not be worth reporting as an effect.

Part 2. Definitions and Equations

Every quantity the studies measure, and every choice inside its definition, with its source. Where a source gives a definition but leaves a parameter free, the parameter is set here and marked.

Table 2.1. Quantities of training and evaluation.

Quantity	Definition and settled choices	Source
Episode	One run of a task from reset to termination or truncation, 1,000 steps.	Ji et al. (2023)
Episodic cost	The sum over an episode of the per-step cost the task returns; cost is incurred on contact with the hazards and other unsafe elements the task defines. Undiscounted.	Ji et al. (2023)
Episodic return	The sum over an episode of the per-step reward. Undiscounted for evaluation; discounted inside learning with OmniSafe's default discount factor, confirmed at the commit.	Ji et al. (2024)
Cost budget d	The largest mean episodic cost allowed: $d = 25$ per episode, the OmniSafe default cost limit for Safety-Gymnasium tasks and one of the two limits Spoor et al. analyse.	OmniSafe documentation; Spoor et al. (2026)

Quantity	Definition and settled choices	Source
Evaluation cost of a checkpoint	Mean episodic cost over 100 evaluation episodes (decision) run with the deterministic policy (the mean of the Gaussian action distribution) on the training task with its default layout distribution. Two disjoint sets of 100 episodes with different evaluation seeds are used: a selection set, run on each of a run's last ten checkpoints and used only to choose one of them in Part 4, and a measurement set, run on the chosen checkpoint only and used for the reported in-distribution cost and as the baseline of every robustness gap. Choosing and measuring on the same episodes would bias the reported cost of the chosen checkpoint downward. The reason for 100 rather than the 50 of Yao et al. (2023): episodic cost on these tasks is heavy-tailed, and the OmniSafe benchmark reports dispersions of the same order as the means, so evaluation noise must be reduced before matching within 2.5 units is meaningful.	Decision; Ji et al. (2024), Appendix B
Final in-distribution cost	The evaluation cost of the checkpoint selected for an arm by the comparison rule of Part 4.	Definition of the study
Constraint onset	The training step at which the cost term enters the surrogate objective (equation 3) and the multiplier begins to update (equation 4). Before onset the cost advantage is not computed into the loss and the multiplier is held at its initial value.	Definition of the study
Onset fraction N	The onset step divided by the total training steps T. Levels: 0, 0.10, 0.25, 0.50.	Definition of the study
Abrupt onset	At onset the budget $d = 25$ applies at once, with the multiplier at its initial value 0.001 (OmniSafe default).	Definition of the study; OmniSafe documentation
Linear ramp	From onset the effective budget in the multiplier update falls linearly from d_{loose} to d over a window $W = 0.10 T$ (decision), then stays at d (equation 5). d_{loose} is twice the episodic cost of an unconstrained PPO agent on the task at 10 million steps as reported in the OmniSafe benchmark: 111 for SafetyPointGoal1-v0 (from 55.72), 116 for SafetyCarGoal1-v0 (from 58.06) and 305 for SafetyPointButton1-v0 (from 152.48). At this value the ramp begins non-binding on every task, which is what a ramp requires. The family of budgets relaxed during training and tightened toward the deployment limit is that of Shperberg et al. (2024).	Decision; Ji et al. (2024), Appendix B; Shperberg et al. (2024)
Robustness gap	The evaluation cost of a matched checkpoint under a battery condition minus its final in-distribution cost (equation 1). One value per seed per condition.	Definition of the study; the conditions are distribution shifts in the sense of Kirk et al. (2023)
Recovery time	Training steps from onset until the training-batch mean episodic cost first falls to or below d . Secondary outcome, late arms only.	Definition of the study

Table 2.2. The robustness battery of Study A.

Condition	Definition and settled choices	Source
Hazard relocation	Hazard positions are re-sampled with the task's own layout generator using twenty new layout seeds, five episodes per layout, 100 episodes in all (decision, matching the evaluation episode count); the policy is not updated.	Ji et al. (2023) layout generator; decision
Dynamics perturbation	Body mass scaled by 1.3 and ground friction by 0.7 in the simulator's model file (decision: a change large enough to alter how actions move the robot and small enough that the task stays solvable, which the reference arm's cost under the perturbation will show); 100 episodes on the training layout distribution; the policy is not updated.	Decision
Reward-only fine-tuning	Training continues for 1,000,000 steps, one tenth of T (decision), with the cost term removed from the objective and the multiplier frozen at zero; evaluation cost is measured after fine-tuning on the training task.	Definition of the study; decision
Transfer	The policy is fine-tuned for 1,000,000 steps, one tenth of T (decision), on the held-out task on the same robot (Goal to Button and Button to Goal) under the same budget d; evaluation cost is measured on the held-out task after fine-tuning.	Definition of the study; decision

Table 2.3. Plasticity metrics of Study A.

Quantity	Definition and settled choices	Source
Plasticity	"The ability of a neural network to quickly change its predictions in response to new information" (Lyle et al., 2023). Its loss during training is documented in continual learning by Dohare et al. (2024) and, in the on-policy regime PPO belongs to, by Juliani and Ash (2024), who also find that remedies developed off-policy can fail on-policy; that caution is why Study A tests its interventions rather than assuming them.	Lyle et al. (2023); Dohare et al. (2024); Juliani and Ash (2024)
Network measured	The actor. Dormant-neuron fraction is computed over all hidden units of the actor; effective rank over the actor's penultimate-layer features; parameter norm over all trainable parameters of the actor, with the two critics logged separately.	Setting; the proposal leaves the network open and this document fixes it
Fixed evaluation batch	2,048 states (decision) collected once per task by a uniform-random policy under seed 0, stored in the repository, and used for every metric computation of every run of that task. Sokar et al. report that the batch size used to detect dormant neurons affects the result, so the batch is fixed and large rather than sampled per run.	Decision; Sokar et al. (2023), appendix
Dormant-neuron score	For neuron i in layer l , the mean absolute activation over the batch divided by the mean of that quantity over all neurons of the layer (equation 6).	Sokar et al. (2023)
Dormant-neuron fraction	The share of hidden neurons whose score is at most $\tau = 0.025$, the threshold Sokar et al. use for ReDo.	Sokar et al. (2023)
Effective rank	The smallest k such that the first k singular values of the penultimate feature matrix on the batch sum to at least $(1 - \delta)$ of all singular values, with $\delta = 0.01$ (equation 7).	Kumar et al. (2021)
Parameter norm	The Euclidean norm of all trainable parameters.	Standard
Logging points	Every 200,000 steps, that is every ten epochs (decision), at onset, and at the end of training.	Decision

Table 2.4. Interventions and controls of Study A.

Quantity	Definition and settled choices	Source
Partial reset	At onset, the last two layers of the actor (the output layer and the hidden layer before it) are reinitialised with the network's default initialiser, and the optimiser state for those parameters is cleared. The critics are not reset. Reset depth of one layer is an ablation.	Nikishin et al. (2022); Setting
Plasticity injection	At onset, the last two layers of the actor are frozen; two fresh copies of those layers are created with identical initialisation; the actor's output becomes the sum in equation (8); only the new copy is trained thereafter. The output is unchanged at the moment of injection.	Nikishin et al. (2023)
Additional constrained training	The late arm continues under the constraint for $N \cdot T$ further steps after T , with no change to the network or optimiser. The data control.	Definition of the study
Warm-started multiplier	At onset the multiplier is set to the value the $N = 0$ arm of the same task and seed had reached at the same step, instead of 0.001.	Definition of the study
Rate-limited multiplier	Each multiplier update is clipped so that the multiplier changes by at most 5 percent of its current value plus 0.01 per epoch (decision: enough to prevent a step response, not enough to prevent convergence within the window the reference arm needs).	Decision
PID multiplier	OmniSafe's PIDLagrangian update (equation 9), using the default gains of OmniSafe's CPPOPID configuration for the pinned version, copied verbatim into Appendix A at the registration commit; run as a further check on SafetyPointGoal1-v0 only.	Stooke et al. (2020); OmniSafe documentation
Multiplier overshoot	The maximum multiplier value in the 2,000,000 steps after onset minus its final value, where the final value is its mean over the last 10 percent of training epochs. OmniSafe's documentation lists overshoot among the known problems of Lagrangian methods, which is the premise of H4.	Åström and Murray (2008); OmniSafe documentation; Setting
Settling time	Steps from onset until the multiplier stays within ± 10 percent of its final value for the rest of training (decision: the 2 and 5 percent bands of control practice are too narrow for a multiplier updated from noisy episode costs).	Åström and Murray (2008); decision
Constrained-steps matched	T is extended for late arms so that every arm trains under the constraint for T steps; a late arm's total is $T/(1 - N)$.	Definition of the study
Total-steps matched	Every arm trains T steps in total; a late arm trains $(1 - N) \cdot T$ steps under the constraint.	Definition of the study

Table 2.5. Definitions of Study B.

Quantity	Definition and settled choices	Source
Constraint level	A cost budget d on the episodic cost of Table 2.1. Stricter means a smaller budget. Levels are budgets, not distances, because Safety-Gymnasium constrains cumulative cost and the prior work varies budgets.	Ji et al. (2023); Yao et al. (2023)
Training set of levels	Single: {10}, {20} or {40}. Sparse: {10, 40}. Moderate: {10, 20, 40}. Dense: {10, 17.5, 25, 32.5, 40}. Continuous: a budget drawn uniformly from [10, 40] at each episode, with one multiplier per 5-unit bin of the range. This reproduces the range-sampled training distribution of Sootla et al. (2022) inside our budget-conditioned PPO-Lagrangian; it is not their Sauté formulation, which augments the state with the remaining allowance and reshapes the cost. Every level lies below the unconstrained PPO cost of 55.72 reported for SafetyPointGoal1-v0 in the OmniSafe benchmark, so every level binds; 10 and 25 are also the limits analysed by Spoor et al. (2026). One unconstrained run in the pilot confirms the binding on the pinned software.	Decision; Ji et al. (2024), Appendix B; Spoor et al. (2026)

Quantity	Definition and settled choices	Source
Unseen budgets	{5, 15, 30, 45} (decision): 5 is a stricter extrapolation, 15 and 30 are interpolations, 45 is a looser extrapolation that still lies below 55.72 and therefore still binds. Distance is defined in equation (11).	Decision; Ji et al. (2024), Appendix B
Budget conditioning	The active budget divided by 100 is appended to the observation vector. During training each episode draws its budget uniformly at random from the arm's training set. One multiplier is kept per training level and is updated only from episodes run at that level, by equation (4). Standard PPO-Lagrangian carries one multiplier and cannot train under several budgets at once; this is the smallest change that lets it.	Setting; the conditioning is that of Yao et al. (2023) in its simplest form
Matched total steps	Every arm trains $T = 10,000,000$ steps regardless of how many levels it contains.	OmniSafe on-policy benchmark (Ji et al., 2024)
Zero-shot evaluation	The unseen budget is placed in the observation and 100 episodes are run with the deterministic policy; no update.	Decision, as in Table 2.1
Few-shot adaptation	One continuation of 1,000,000 steps per unseen budget, under that budget with a fresh multiplier at 0.001; satisfaction is measured at 200,000, 500,000 and 1,000,000 steps (decision: 2, 5 and 10 percent of T). The three horizons are read from one continuation, not three.	Decision
Constraint-satisfaction rate	The fraction of evaluation episodes whose episodic cost is at most the unseen budget (equation 10).	Definition of the study
Violation magnitude	The mean over evaluation episodes of $\max(\text{cost} - \text{budget}, 0)$.	Definition of the study
Adaptation steps	The smallest few-shot horizon at which the satisfaction rate reaches 0.80 (decision); recorded as above the largest horizon if never reached.	Decision

2.5 Equations used in this document

Symbols: C denotes an evaluation cost, with C_{ID} the in-distribution cost and C_{cond} the cost under a battery condition; d the cost budget, λ the Lagrange multiplier, η its learning rate, J_C the mean episode cost of the current epoch, A_R and A_C the reward and cost advantages, N the onset fraction, T the total steps, W the ramp window.

Robustness gap of one seed under one condition. The cost under the condition minus the in-distribution cost of the same matched checkpoint.

$$gap = C_{cond} - C_{ID} \quad (1)$$

Gap difference between a late arm and the reference arm. The estimand of H1; positive when late onset is worse.

$$\Delta(N) = gap(N) - gap(0) \quad (2)$$

PPO-Lagrangian surrogate objective. The advantage the policy maximises once the constraint is active; before onset only A_R is used.

$$A = \frac{A_R - \lambda A_C}{1 + \lambda} \quad (3)$$

Multiplier update, once per epoch. Integral control on the constraint violation: the multiplier accumulates the excess of cost over budget and cannot go below zero. This is the update written in OmniSafe's own documentation of Lagrangian methods, applied there through the optimiser named in the configuration.

$$\lambda \leftarrow \max(0, \lambda + \eta(J_C - d)) \quad (4)$$

Ramped effective budget. For a step t after onset at $N \cdot T$; after the window the budget stays at d .

$$d_{eff}(t) = d_{loose} - (d_{loose} - d) \cdot \frac{t - NT}{W}, \quad NT \leq t < NT + W \quad (5)$$

Dormant-neuron score. Mean absolute activation of neuron i in layer l over the fixed batch, divided by the layer's mean; a neuron is dormant if the score is at most $\tau = 0.025$.

$$s_i^l = \frac{E |h_i^l(x)|}{\frac{1}{Hl} \sum_{k=1}^H E |h_k^l(x)|} \quad (6)$$

Effective rank of the penultimate feature matrix Φ . The fewest of the r singular values σ , in decreasing order, that carry a share $1 - \delta$ of the spectrum, with $\delta = 0.01$.

$$sr\text{rank}_\delta(\Phi) = \min \left\{ k : \frac{\sum_{i=1}^k \sigma_i}{\sum_{i=1}^r \sigma_i} \geq 1 - \delta \right\} \quad (7)$$

Plasticity injection. The frozen head keeps the learned function; the new trained head and its frozen copy cancel at the moment of injection and diverge as training continues.

$$f(x) = f_\theta(x) + f_{\theta_1}(x) - f_{\theta_2}(x) \quad (8)$$

PID multiplier update (OmniSafe form). The error e is the epoch's mean episode cost minus the budget; the integral term is clamped at zero; gains are the pinned CPPPID defaults.

$$\lambda_{t+1} = \max \left(0, K_P e + K_I \int e dt + K_D \frac{de}{dt} \right) \quad (9)$$

Constraint-satisfaction rate of Study B. Over $n = 100$ evaluation episodes under an unseen budget; the indicator is 1 when the episode's cost is within the budget.

$$SR(d_{test}) = \frac{1}{n} \sum_{e=1}^n \mathbb{1}[C_e \leq d_{test}] \quad (10)$$

Distance of an unseen budget from the training set. Used to order the unseen budgets and to separate interpolation from extrapolation.

$$\text{dist}(d_{test}) = \min_{d \in \text{training set}} |d_{test} - d| \quad (11)$$

Standardised difference and the go-signal bound. Cohen's d with the pooled seed standard deviation s ; and the pilot's upper bound with three seeds, which uses the t distribution with two degrees of freedom rather than a bootstrap, because three values cannot support one.

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \quad U_{80} = \bar{x} + 1.886 \cdot \frac{s}{\sqrt{3}} \quad (12)$$

Part 3. Design

3.1 Environment and software, both studies

Table 3.1. Environment, software and hardware.

Item	Setting	Source
Tasks	Study A: SafetyPointGoal1-v0, SafetyCarGoal1-v0, SafetyPointButton1-v0, with SafetyPointGoal1-v0 as the primary task. Study B: SafetyPointGoal1-v0 only.	Ji et al. (2023)
Algorithm	PPO-Lagrangian as implemented in OmniSafe: surrogate of equation (3); multiplier updated once per epoch by equation (4), which is the update OmniSafe's own documentation gives, with learning rate 0.035 from initial value 0.001 and cost limit 25; 500 epochs of 20,000 steps, the configuration OmniSafe's documentation itself uses for SafetyPointGoal1-v0. Three further settings, 40 update iterations per epoch with minibatches of 64 and a target KL of 0.02, are taken from a published replication of OmniSafe's defaults and are confirmed against the pinned configuration at the commit. Every remaining hyperparameter (learning rates, discount factors, GAE parameters, clipping) is OmniSafe's default for the pinned version and is copied verbatim into Appendix A.	OmniSafe documentation (Ji et al., 2024); replication for three settings

Item	Setting	Source
Networks	Actor, reward critic and cost critic are multilayer perceptrons with two hidden layers of 64 units and tanh activation, OmniSafe's documented defaults for PPO-Lagrangian. Study B appends one input, the normalised budget, to the actor and critic inputs.	OmniSafe documentation, Lagrangian methods page
Total steps T	10,000,000 per run: 500 epochs of 20,000 steps, the setting of OmniSafe's on-policy benchmark, at which its published results for these tasks were obtained.	OmniSafe on-policy benchmark (Ji et al., 2024)
Software versions	Safety-Gymnasium, OmniSafe, MuJoCo, PyTorch and Python versions are pinned in the repository's environment file at the registration commit and recorded in Appendix A.	Setting
Hardware and device	One workstation; specification recorded in Appendix A. Runs execute on CPU with one torch thread, one vectorised environment and one process each, as OmniSafe recommends for consistency with its published results; the number of runs executed concurrently is set by the core count and measured in the pilot [P].	OmniSafe on-policy benchmark (Ji et al., 2024); decision
Determinism check	Two runs with the same seed and configuration must produce identical evaluation cost at the first checkpoint before any arm is launched.	Setting

3.2 Study A: factors and arms

Table 3.2. Factors of the main sweep.

Factor	Levels	Purpose
Onset fraction N	0, 0.10, 0.25, 0.50	The manipulation of H1
Onset shape	Abrupt; linear ramp (d_loose per task as in Table 2.1, $W = 0.10 T$)	The manipulation of H2
Step-matching control	Constrained-steps matched; total-steps matched	Removes unequal constrained training as an explanation
Task	Three, as in Table 3.1	Generality across robot and task
Seeds	Five per arm: 0, 1, 2, 3, 4. The seed sets network initialisation, layout sampling, action sampling and minibatch order.	Uncertainty in every estimate

The $N = 0$ arm has no onset shape and no step-matching variant, so each task has $1 + 3 \times 2 \times 2 = 13$ main arms; with five seeds and three tasks the main sweep is 195 runs.

Table 3.3. Additional arms of Study A.

Group	Arms	Runs	Cut order
Mediation treatments (H3)	Reset, injection, additional constrained training, applied at onset to the abrupt arm under the total-steps-matched control, at $N = 0.10, 0.25, 0.50$, in three tasks	$3 \times 3 \times 3 \times 5 = 135$	Fourth
Controller variants (H4)	Warm-started and rate-limited multiplier, abrupt arms, total-steps matched, $N = 0.10, 0.25, 0.50$, three tasks	$2 \times 3 \times 3 \times 5 = 90$	Fourth
PID check (H4)	PID multiplier, abrupt arms, SafetyPointGoal1-v0 only, $N = 0.10, 0.25, 0.50$	$3 \times 5 = 15$	First
Ablations	Ramp window $W \in \{0.05, 0.20\} T$; reset depth one layer; injection on the last layer only; budget $d \in \{10, 50\}$; $T = 5,000,000$. SafetyPointGoal1-v0, $N = 0.50$ only.	Set by amendment after the go decision	Exploratory unless registered by amendment before they run

3.3 Study B: arms and evaluation

Table 3.4. Arms of Study B, all on SafetyPointGoal1-v0, five seeds each.

Arm	Training levels	Role
Single-10	{10}	Single-level baseline; nearest to the unseen budgets 5 and 15
Single-20	{20}	Single-level baseline; nearest to 15 and 30
Single-40	{40}	Single-level baseline; nearest to 30 and 45
Sparse	{10, 40}	Two levels spanning the range
Moderate	{10, 20, 40}	Three levels spanning the range
Dense	{10, 17.5, 25, 32.5, 40}	Five levels spanning the range
Continuous	Uniform on [10, 40] per episode	Range sampling as in Sootla et al. (2022), within our algorithm; the upper reference for coverage

Seven arms with five seeds are 35 training runs. Each finished run is evaluated zero-shot on the four unseen budgets, then continued once under each unseen budget for 1,000,000 steps, with satisfaction read at the three horizons of Table 2.5: $35 \times 4 = 140$ continuations of 1,000,000 steps, equivalent to 14 full runs. Study B's total is therefore 49 run-equivalents. Study B's runs are scheduled after Study A's main sweep and are cut, if compute runs short, in the order of Part 6.1; its zero-shot arms are never cut.

3.4 Checkpoints and logging

Each run saves a checkpoint every 200,000 steps, at onset, and at the end of training, and logs at each checkpoint: training-batch mean episodic cost and return, the three plasticity metrics on the fixed batch (Study A), and, for the final ten checkpoints, the selection-set cost and return over 100 episodes; the selected checkpoint is then evaluated on the 100-episode measurement set. The multiplier, or in Study B every per-level multiplier, is logged at every epoch. Every run writes one row to the results ledger (Appendix B) at completion, including the commit hash, the configuration file hash, the seed, the wall-clock time and the machine.

3.5 Reference values from the published benchmark

The OmniSafe benchmark (Ji et al., 2024, Appendix B) reports, at 10 million steps with cost limit 25, the following means and dispersions over its evaluation iterations and seeds. The unconstrained values are identical in the first and latest arXiv versions of the paper; the Lagrangian values are from the latest. They are the sources for d_loose, for Study B's budget levels, and for the expectation that the budget binds on every task.

Table 3.5. Published reference values at 10 million steps, cost limit 25 (Ji et al., 2024, Appendix B, benchmark tables 3a and 3b of the arXiv version).

Task	Unconstrained PPO reward	Unconstrained PPO cost	Lagrangian methods, reward and cost
SafetyPointGoal1-v0	25.44 ± 5.43	55.72 ± 35.55	PPO-Lagrangian cost 25.80 ± 34.99 (reward cell not legible in the extraction; copied at the commit); TRPO-Lagrangian 18.51 ± 3.83 and 22.98 ± 8.45 ; RCPO 15.27 ± 4.05 and 30.56 ± 19.15
SafetyCarGoal1-v0	33.41 ± 2.89	58.06 ± 42.06	PPO-Lagrangian 13.27 ± 9.26 and 21.72 ± 32.06
SafetyPointButton1-v0	27.03 ± 6.14	152.48 ± 80.39	PPO-Lagrangian 4.60 ± 4.73 and 20.80 ± 35.78 ; TRPO-Lagrangian 6.93 ± 1.84 and 31.16 ± 20.58

Two consequences follow. First, an unconstrained agent exceeds the budget of 25 on every task by a wide margin, so the constraint binds and a late onset has something to change. Second, the dispersions are of the same order as the means, and on the primary task PPO-Lagrangian itself finishes at a mean cost of 25.80 with a dispersion of 34.99; the benchmark pools evaluation iterations and seeds in its dispersion, so it overstates

the seed-to-seed spread, but it is the reason the evaluation episode count is 100 and the reason G1 in Part 6 checks matching feasibility before anything else.

3.6 The pilot

The pilot: eight runs, completed before the go decision.

Study A: SafetyPointGoal1-v0; arms $N = 0$ and $N = 0.50$, abrupt, total-steps matched; seeds 0, 1, 2; battery conditions hazard relocation and dynamics perturbation. Six runs.

Study B: one unconstrained PPO run, to confirm that every budget of Table 2.5 lies below the unconstrained cost on the pinned software; one Moderate arm run with seed 0, evaluated with each of its three training budgets in the observation, to confirm that its mean costs are ordered with the budgets and each within 2.5 of its budget (G4). Two runs.

Measured: wall-clock time per run; runs sustained concurrently; seed-to-seed standard deviation of final in-distribution cost per arm; $\Delta(0.50)$ under hazard relocation over three seeds; the reference arm's cost under the dynamics perturbation; the unconstrained cost; the Moderate arm's satisfaction rates. These are the [P] quantities of Part 8. The pilot's runs are not reused. It decides, by Part 6, whether the studies proceed.

Part 4. Comparison Rule

4.1 Study A: matching on in-distribution cost

1. Selection, applied to every arm including the reference. For each seed of each arm, the checkpoint is chosen from the run's last ten checkpoints (the final 2,000,000 steps; decision) as the one whose selection-set cost is closest to the budget $d = 25$. The target is the budget, not another arm's cost, so that the rule treats every arm identically and no arm is selected toward another. The step index of the chosen checkpoint is recorded in the ledger as its training age.
2. Reference. For each task and step-matching control, the reference cost is the mean over the five seeds of the $N = 0$ arm's selected checkpoints.
3. Feasibility. The reference arm must satisfy the budget: reference cost $\leq d + \text{tolerance}$. If it does not, the task is reported as infeasible at $d = 25$ and enters no comparison; the design is not adjusted to rescue it.
4. Matched cost. An arm's matched cost is the mean over its five seeds of the selected checkpoints' evaluation costs.
5. Tolerance. An arm is matched if $|\text{matched cost} - \text{reference cost}| \leq 2.5$ cost units, one tenth of the budget (decision). The tolerance is not widened for any arm. If the reference arm's own seed-to-seed standard deviation of selected-checkpoint cost exceeds 10 cost units in a task, matching in that task is reported as infeasible and the task contributes no comparison; that condition is checked in the pilot (G1).
6. Unmatched arms. An arm that cannot be matched is reported with its cost and left out of the battery. It is not compared, and the tolerance is not widened for it. A late arm that never reaches the budget is a finding about learning, reported as such, and not a finding about robustness.
7. Sensitivity. The primary analysis is repeated twice and reported in the paper's appendix, labelled exploratory: once with tolerance 5.0, and once with every arm's final checkpoint in place of the selected one, which shows how much the selection itself moved the result.

4.1.1 The two estimands and what the matched one can and cannot say

The matched-checkpoint estimand (primary). The difference in robustness gap between onset conditions among policies that reached the same in-distribution cost. It is a comparison conditional on cost achievement, not a randomised contrast: onset timing is assigned by us, but which checkpoint of a run enters the comparison is chosen by an observed, noisy quantity. The estimand therefore answers the practitioner's question, "given two policies that pass the same safety check, does it matter when the constraint arrived?", and does not by itself isolate onset timing from everything that co-varies with reaching the budget, such as the training age of the selected checkpoint or the state of the multiplier at that point.

The final-checkpoint estimand (sensitivity). The difference in robustness gap at the end of training, whatever the in-distribution cost. It is the unconditional contrast and is reported beside the primary one; agreement between the two strengthens the causal reading, disagreement limits it.

What is reported with both. The training age of every selected checkpoint, its selection-set and measurement-set costs, and the multiplier value at selection, so that a reader can see how far the matched checkpoints differ in anything other than onset timing. Where selected training ages differ systematically between arms, that difference is named in the paper as a limit on the causal interpretation.

4.2 Study B: the comparison each hypothesis uses

- G1 compares Sparse, Moderate and Dense with one another on the mean satisfaction rate over the four unseen budgets; G5 compares Dense with Continuous on the same outcome.
- G2 compares Dense with the single-level arm nearest to each unseen budget by equation (11). Budget 5 has one nearest arm, Single-10; budget 45 has one, Single-40. Budgets 15 and 30 are each equidistant from two single-level arms; for those the comparator is whichever of the two has the higher satisfaction on that budget, the choice least favourable to G2, and both are reported. G2 never compares Dense only with Single-10.
- G3 compares, within each arm, the drop in satisfaction at budget 5 with the drop at budget 45, each drop measured from the arm's satisfaction at its own nearest training level.
- G4 compares the adaptation steps of Sparse, Moderate and Dense on each unseen budget, with adaptation steps censored at the largest horizon when the target is never reached.
- Floor rule, for every comparison above. If more than half of the arms have a satisfaction rate below 5 percent on an unseen budget, that budget's comparison is made on violation magnitude (Table 2.5) instead of satisfaction rate, and the report says so; a rate at the floor cannot separate arms.

Part 5. Analysis

5.1 Unit, structure, and estimands

The unit of analysis is the seed in both studies, and every interval is computed by treating the seed as the only independent replicate. Within a seed, the 100 evaluation episodes, the four battery conditions, and in Study B the four unseen budgets are repeated measures; they are summarised into one number per seed per outcome before any interval is computed, and are never resampled as if they were independent. Where two arms share the same seed list, seed k of one arm and seed k of the other share their network initialisation and layout sequence and are treated as a pair; paired and unpaired analyses are both reported, the unpaired as primary because five pairs give too few degrees of freedom for pairing to help unless the seed correlation is strong. Study A: for each task, control, battery condition and arm, the outcome per seed is the robustness gap of its matched checkpoint (equation 1); the estimands are differences between arms in the mean gap over seeds, equation (2) for H1, $\text{gap}(\text{abrupt}) - \text{gap}(\text{ramp})$ at each N for H2, and $\text{gap}(\text{treated}) - \text{gap}(\text{untreated})$ at $N = 0.50$ for H3 and H4. Study B: for each arm and unseen budget, the outcome per seed is the satisfaction rate of

equation (10), zero-shot and after each few-shot horizon, and the adaptation steps; the estimands are the differences named in Part 4.2.

5.2 Primary and secondary outcomes

Study A primary outcome: the robustness gap under hazard relocation on SafetyPointGoal1-v0. The primary analysis is H1 on that outcome under both step-matching controls.

Study B primary outcome: the mean zero-shot satisfaction rate over the four unseen budgets, one number per seed. The primary analysis is G1 and G2 on that outcome, each tested at $\alpha = 0.025$ (Bonferroni within the primary family of two), so that for G1 and G2 an interval that must exclude zero is the 97.5 percent interval.

Everything else is secondary: the other three battery conditions, the other two tasks, H2 to H4, recovery time, return, the plasticity correlations, G3, G4, G5, satisfaction per budget, violation magnitude. A claim about a secondary outcome is corrected for multiplicity within its family (the four battery conditions form one family; the three tasks another; the four unseen budgets another) by the Holm procedure at $\alpha = 0.05$.

5.3 Estimation and intervals

Each estimand is reported as the difference of seed means, as Cohen's d with the pooled seed standard deviation (equation 12; Cohen, 1988), and with two 95 percent intervals: a two-sample t -interval with Welch's degrees of freedom (Welch, 1947), which is the primary interval because with five seeds per arm a percentile bootstrap has too few distinct resamples to hold its nominal coverage; and a percentile bootstrap interval (Efron and Tibshirani, 1993) from 10,000 resamples of seed indices with replacement, reported beside it. The resampling unit is the seed index: a resample takes every outcome of the chosen seeds together, so the paired and repeated-measure structure is preserved. Where a hypothesis box says an interval excludes zero, it means the primary interval. Following Agarwal et al. (2021), the interquartile mean with a stratified bootstrap over seeds and evaluation episodes is reported alongside the mean. Where a hypothesis box says an interval excludes zero, it means the 95 percent percentile interval over seeds.

5.4 Trend and correlation tests

The increase of Δ with N (H1) and of satisfaction with coverage (G1) are tested by Spearman's rank correlation between the factor level and the seed-level outcome (for H1, twenty points: four onset fractions by five seeds), with an interval from resampling seed indices, each resample carrying all of a seed's factor levels together; the ordering condition holds if the correlation is positive with an interval excluding zero. The plasticity correlations of H3(a) use Spearman's correlation between each metric at onset and the seed-level gap across all late arms of a task, with the same interval.

5.5 Power

With five seeds per arm, a two-sample t -test at $\alpha = 0.05$ (two-sided) has 80 percent power at a standardised difference of 2.02, 55 percent at 1.5 and 29 percent at 1.0; with eight seeds the 80-percent point falls to 1.51, with ten to 1.33, with twelve to 1.20 (all from the noncentral t distribution). Differences below the detectable size are reported with their intervals and are not claimed as findings; each study can establish a large effect and can bound a small one, and at five seeds cannot resolve a medium one. That is the reason the seed count is the last item in the cut order and is never reduced, and the reason for the rule below.

Surplus compute goes to seeds first. If the pilot's throughput measurement (G2) shows capacity beyond the 484 registered run-equivalents, the surplus is spent, before anything else is added, on additional seeds for the arms that carry the primary comparison: the $N = 0$ and $N = 0.50$ arms under both step-matching controls on SafetyPointGoal1-v0, and the seven arms of Study B. Seeds are added in the order 5, 6, 7, and so on, the same count for every arm in the set, up to twelve. The seed count actually used is recorded in Part 8 at the go decision, and the power statement in the paper is computed from it.

5.6 Exclusion rule

A run is excluded only if it crashes, fails to complete its steps, or produces a non-finite loss or multiplier; the exclusion and its cause are recorded in the ledger and the run is repeated with the next unused seed (5, 6, and so on). A run that completes is kept whatever its numbers. No run is excluded on the basis of its results.

5.7 Reporting a null result

If H1 or G1 is falsified, the result is reported as an upper bound: the upper limit of the 95 percent interval of the primary estimand. The statement 'no effect' is not used. A bound below the minimum effect of interest is reported as evidence that any effect is smaller than practical relevance in this setting; a bound above it is reported as inconclusive at this sample size.

5.8 Analysis code

The analysis script is committed to the repository before the first main-study run completes, is run once on the pilot data to test it, and is not edited between the go decision and the final analysis except to fix errors, each of which is recorded in Part 9.

Part 6. Go or No-Go Rule and Cut Order

The decision is taken by the group at the go meeting, from the pilot's results and nothing else. Four conditions are checked in order.

G1, feasibility (Study A). Both pilot arms reach a final in-distribution cost (Table 2.1: the cost of the checkpoint selected by Part 4) of at most $d + 2.5$ within T ; the reference arm's seed-to-seed standard deviation of that cost is at most 10, with the same statistic for the final checkpoints reported beside it; and the two arms can be matched under the tolerance of Part 4.

G2, throughput (both studies). The measured wall-clock time per run, divided by the measured number of concurrent runs, multiplied by 484 run-equivalents (435 for Study A, 49 for Study B), is at most the machine-hours the group allocates to the registered runs. The allocation is written into Part 8 before the pilot's throughput is read, so that it cannot be set to fit the result.

G3, signal (Study A). The pilot's mean $\Delta(0.50)$ under hazard relocation over the three seeds is at least 5 cost units, or its upper bound U_{80} of equation (12) is at least 5.

G4, conditioning (Study B). Every budget of Table 2.5 lies below the pilot's unconstrained cost; and the Moderate arm, evaluated with each of its three training budgets in the observation, shows mean costs that are ordered with the budgets (lowest at 10, highest at 40) and each at most its budget plus 2.5. The ordering is the test that the policy reads the budget input; the level is the test that the per-level multipliers work.

Go if all four hold: the measured values are entered in Part 8, the amended document is committed, and Parts 1, 4, 5 and 6 are frozen. **Revise once** if G1, G2 or G4 fails: one revision of T , d , W , the tolerance, or Study B's conditioning is permitted, recorded in Part 9, followed by a re-pilot of the same size; a second failure of the same condition is a no-go for the study it concerns. **No-go for Study A** if G1 and G2 hold and G3 fails: any timing effect is below the minimum effect of interest in the pilot setting; the pilot is written up as a negative pilot with its bound, and Study B becomes the main study.

6.1 Cut order if G2 fails after the revision

Cuts are made in this order, each applied only if the previous is insufficient.

1. Drop the PID check (15 runs).
2. Drop SafetyCarGoal1-v0 from every group of Study A (65 main-sweep runs, 45 treatment runs, 30 controller runs).
3. Shorten Study B's few-shot continuations from 1,000,000 to 200,000 steps, keeping only the first horizon; the three horizons are read from one continuation, so this, not dropping horizons, is what saves compute (about 11 run-equivalents).
4. Restrict Study A's mediation treatments and controller variants to $N = 0.50$.
5. Drop $N = 0.10$ from the main sweep (4 arms per remaining task, 20 runs per task). It contributes to the trend test of $H1$ and not to the primary comparison, which is $N = 0.50$ against $N = 0$.

Never cut: the number of seeds; Study B's seven zero-shot arms; the $N = 0$ and $N = 0.50$ arms under both step-matching controls on SafetyPointGoal1-v0 and SafetyPointButton1-v0, which carry the primary comparison.

Part 7. What Is New in Each Study

7.1 Study A

Shperberg et al. (2024) relax the cost limit during training and tighten it toward the deployment limit, and show that the deployment limit is met with higher return; their measurement is taken at the end of training in the training distribution. Koprulu et al. (2025) order tasks by safety rather than timing the constraint. In language models, Qi et al. (2024, 2025) show that safety added after capability training can be shallow and is undone by benign fine-tuning, and Sam et al. (2026) show that when safety enters pretraining changes its robustness; none of those settings has a cost budget or a measurable representational mechanism. What is new in Study A is the combination: an onset sweep with in-distribution cost matched, a robustness battery, a

mechanism test with a data control and a manipulation check, and a controller analysis that can attribute the effect elsewhere. The closest prior results are cited in the paper with the sentence in which each stops.

7.2 Study B

Two prior results bound what Study B can claim. Yao et al. (2023) show that a constraint-conditioned policy trained on three cost budgets (20, 40 and 60) generalises zero-shot to unseen budgets across a wider range, inside and outside the training range, with a smooth performance curve; their appendix examines the choice of behaviour policies, meaning the choice of training budgets. Sootla et al. (2022), with the budget entering the state as the remaining safety allowance, train policies at a single budget of 60 and evaluate them at 40 and 80, and separately train policies with the budget sampled uniformly from 5 to 100; on the unseen budgets the range-trained policies satisfy the constraint better than the single-budget ones, over five seeds. Testing an unseen budget, sweeping the distance to the nearest training budget, separating interpolation from extrapolation, and showing that training on a range beats training at one budget are therefore not contributions of Study B.

Study B claims three things and no others:

- (1) the number and spacing of discrete training budgets, varied as the independent variable with a fixed standard algorithm and matched total steps, and compared with the continuous range that Sootla et al. used (G1, G5);
- (2) a comparison against the single-level agent nearest to each unseen budget, with single-level agents at three budgets rather than one, which removes proximity as the explanation of any advantage (G2);
- (3) the few-shot adaptation cost as a function of coverage, and the asymmetry between stricter and looser unseen budgets (G3, G4).

A targeted search for prior work that varies the number or spacing of training budgets as the independent variable is run by the protocol of Appendix C and its record is committed to the repository before Study B's first run; the path of the record is entered in Table 9.0. If it finds such a paper, Study B is withdrawn by amendment and the finding is recorded in Part 9. Study B's results are reported in the same paper as Study A if space permits and otherwise separately; the choice does not affect the analysis.

Part 8. Empirical Values and Recorded Decisions

8.1 Quantities supplied by the pilot

Only quantities that no source and no decision can supply are listed here. Each is measured in the pilot and entered at the go decision. Everything else in this document is already fixed by a source or a recorded decision and does not wait for the pilot.

Table 8.1. Empirical quantities measured by the pilot.

Quantity	Why only the pilot can supply it	Used by	Measured value / date
Wall-clock time per run	Depends on the workstation	G2	
Runs sustained concurrently	Depends on the core count and memory	G2	
Machine-hours allocated to the registered runs	A group decision, recorded before the throughput measurement is read	G2	
Seeds per primary-comparison arm actually used	Five, or more by the surplus rule of Part 5.5	Part 5.5; the power statement of the paper	
Seed-to-seed SD of final in-distribution cost, per arm	Depends on the pinned software and the task	G1; the infeasibility threshold of Part 4	

Quantity	Why only the pilot can supply it	Used by	Measured value / date
$\Delta(0.50)$ under hazard relocation, three seeds	The effect itself	G3	
Reference arm's cost under the dynamics perturbation	Shows whether the perturbed task remains solvable	Table 2.2 decision check	
Unconstrained cost on the pinned software	Confirms every Study B budget binds	G4	
Moderate arm's mean cost at each of its training budgets	Confirms the policy reads the budget input and the multipliers hold the level	G4	

8.2 Decisions recorded in this document

Settings fixed by group decision rather than by a published source. Each may be changed only by an amendment in Part 9; a change made after data are in view labels every affected result exploratory.

Table 8.2. Register of decisions.

Setting	Value	Reason given in
Minimum effects of interest	5 cost units (A); 5 percentage points (B)	Part 1.5
Evaluation episodes per checkpoint	100	Table 2.1
Ramp window W; rule for d_loose	0.10 T; twice the unconstrained cost per task	Table 2.1
Hazard-relocation layouts	20 layouts $\times$ 5 episodes	Table 2.2
Dynamics perturbation factors	Mass 1.3; friction 0.7	Table 2.2
Fine-tuning and transfer horizons	1,000,000 steps	Table 2.2
Fixed evaluation batch	2,048 states	Table 2.3
Logging interval	200,000 steps	Table 2.3
Rate limit on the multiplier	5 percent + 0.01 per epoch	Table 2.4
Settling band	± 10 percent of final value	Table 2.4
Study B budgets and conditioning	Training 10 to 40 in seven arms including a continuous range; unseen 5, 15, 30, 45; budget/100 in the observation; one multiplier per level or per 5-unit bin	Table 2.5
Study B few-shot horizons; target	0.2, 0.5, 1.0 million steps; 0.80	Table 2.5
Device and parallelism	CPU; one thread; one environment; one process	Table 3.1
Checkpoint selection, window and tolerance	Closest to d = 25 among the last ten checkpoints, every arm alike; 2.5; infeasible above SD 10	Part 4.1
Primary outcomes	Hazard-relocation gap on SafetyPointGoal1-v0 (A); mean zero-shot satisfaction (B)	Part 5.2
Cut order	As listed	Part 6.1

Part 9. Registration Record and Amendment Log

9.1 Registration record

Table 9.0. Fields that complete the registration; empty until the commit.

Field	Value
Repository	github.com/paf-iaist-ai-research/when-safety-comes-late (public)

Field	Value
Registration commit hash	(entered at commit)
Commit timestamp	(entered at commit)
Tag	registration
Path of the targeted-search record for Study B (Appendix C)	(entered when the search is run)

9.2 Amendment log

Table 9.1. Amendments.

Amendment	Date	Commit hash	Change	Reason	Approved by
Initial registration	(commit timestamp)	(entered at commit)	Registration of Study A and Study B. Replaces earlier drafts: settings fixed from OmniSafe's published benchmark and defaults or recorded as decisions; Study B registered as a tested study with hypotheses G1 to G4; the go rule's signal condition uses a t-interval; power computed.		Group
First amendment	(go decision)		Pilot measurements entered in Part 8; go or no-go recorded; Parts 1, 4, 5, 6 frozen	Pilot results	Group

Part 10. Roles and Review

Table 10.1. Authorship and review of each part.

Part of this document	Written by	Read aloud and checked at the group's review
Definitions and equations (Part 2)	Metrics and interventions owner	Yes / No
Design and comparison rule (Parts 3, 4)	Environment and tests owner	Yes / No
Analysis (Part 5)	Analysis and results owner	Yes / No
Hypotheses and go or no-go (Parts 1, 6)	Pilot owner	Yes / No
Assembly, Part 7, references, reading list	Pre-registration and literature owner	Yes / No

For every number in this document the group asked one question at the review: where does it come from? Every number now carries a source, a recorded decision, or a [P] that names the pilot measurement supplying it. The group is the authority for this document: it adopts it, amends it through Part 9, and takes the go decision.

References

- Agarwal, R., Schwarzer, M., Castro, P. S., Courville, A., & Bellemare, M. G. (2021). Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*. <https://arxiv.org/abs/2108.13264>
- Åström, K. J., & Murray, R. M. (2008). *Feedback systems: An introduction for scientists and engineers*. Princeton University Press.

- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates.
- Dohare, S., Hernandez-Garcia, J. F., Lan, Q., Rahman, P., Mahmood, A. R., & Sutton, R. S. (2024). Loss of plasticity in deep continual learning. *Nature*, 632(8026), 768–774. <https://doi.org/10.1038/s41586-024-07711-7>
- Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman & Hall.
- Ji, J., Zhang, B., Zhou, J., Pan, X., Huang, W., Sun, R., Geng, Y., Zhong, Y., Dai, J., & Yang, Y. (2023). Safety-Gymnasium: A unified safe reinforcement learning benchmark. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*. <https://openreview.net/forum?id=WZmlxluIGR>
- Ji, J., Zhou, J., Zhang, B., Dai, J., Pan, X., Sun, R., Huang, W., Geng, Y., Liu, M., & Yang, Y. (2024). OmniSafe: An infrastructure for accelerating safe reinforcement learning research. *Journal of Machine Learning Research*, 25(285), 1–6. <http://jmlr.org/papers/v25/23-0681.html>
- Juliani, A., & Ash, J. T. (2024). A study of plasticity loss in on-policy deep reinforcement learning. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. <https://arxiv.org/abs/2405.19153>
- Kirk, R., Zhang, A., Grefenstette, E., & Rocktäschel, T. (2023). A survey of zero-shot generalisation in deep reinforcement learning. *Journal of Artificial Intelligence Research*, 76, 201–264. <https://doi.org/10.1613/jair.1.14174>
- Koprulu, C., Simão, T. D., Jansen, N., & Topcu, U. (2025). Safety-prioritizing curricula for constrained reinforcement learning. In *Proceedings of the Thirteenth International Conference on Learning Representations (ICLR 2025)* (pp. 6138–6166).
- Kumar, A., Agarwal, R., Ghosh, D., & Levine, S. (2021). Implicit under-parameterization inhibits data-efficient deep reinforcement learning. In *International Conference on Learning Representations (ICLR 2021)*. <https://openreview.net/forum?id=O9bnihsFFXU>
- Lyle, C., Zheng, Z., Nikishin, E., Avila Pires, B., Pascanu, R., & Dabney, W. (2023). Understanding plasticity in neural networks. In *Proceedings of the 40th International Conference on Machine Learning (PMLR 202, pp. 23190–23211)*. <https://proceedings.mlr.press/v202/lyle23b.html>
- Nikishin, E., Oh, J., Ostrovski, G., Lyle, C., Pascanu, R., Dabney, W., & Barreto, A. (2023). Deep reinforcement learning with plasticity injection. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. <https://arxiv.org/abs/2305.15555>
- Nikishin, E., Schwarzer, M., D’Oro, P., Bacon, P.-L., & Courville, A. (2022). The primacy bias in deep reinforcement learning. In *Proceedings of the 39th International Conference on Machine Learning (PMLR 162, pp. 16828–16847)*. <https://proceedings.mlr.press/v162/nikishin22a.html>
- OmniSafe Team. (n.d.). OmniSafe documentation: Lagrangian methods; Lagrange multiplier; on-policy benchmark. <https://omnisafe.readthedocs.io/en/latest/>
- Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11), 2600–2606. <https://doi.org/10.1073/pnas.1708274114>
- Qi, X., Panda, A., Lyu, K., Ma, X., Roy, S., Beirami, A., Mittal, P., & Henderson, P. (2025). Safety alignment should be made more than just a few tokens deep. In *Proceedings of the Thirteenth International Conference on Learning Representations (ICLR 2025)*. <https://openreview.net/forum?id=6Mxhg9PtDE>
- Qi, X., Zeng, Y., Xie, T., Chen, P.-Y., Jia, R., Mittal, P., & Henderson, P. (2024). Fine-tuning aligned language models compromises safety, even when users do not intend to! In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR 2024)*. <https://openreview.net/forum?id=hTEGyKf0dZ>
- Sam, D., Goyal, S., Maini, P., Robey, A., & Kolter, J. Z. (2026). When should we introduce safety interventions during pretraining? [Preprint]. *arXiv*. <https://arxiv.org/abs/2601.07087>
- Shperberg, S. S., Liu, B., & Stone, P. (2024). Relaxed exploration constrained reinforcement learning. In *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2024)*. <https://dl.acm.org/doi/10.5555/3635637.3663034>
- Sokar, G., Agarwal, R., Castro, P. S., & Evci, U. (2023). The dormant neuron phenomenon in deep reinforcement learning. In *Proceedings of the 40th International Conference on Machine Learning (PMLR 202, pp. 32145–32168)*. <https://proceedings.mlr.press/v202/sokar23a.html>
- Spoor, L., Serra-Gómez, Á., Plaat, A., & Moerland, T. (2026). Towards a practical understanding of Lagrangian methods in safe reinforcement learning [Preprint]. *arXiv*. <https://arxiv.org/abs/2510.17564>
- Sootla, A., Cowen-Rivers, A. I., Jafferjee, T., Wang, Z., Mguni, D. H., Wang, J., & Ammar, H. (2022). Sauté RL: Almost surely safe reinforcement learning using state augmentation. In *Proceedings of the 39th International Conference on Machine Learning (PMLR 162, pp. 20423–20443)*. <https://proceedings.mlr.press/v162/sootla22a.html>

- Stooke, A., Achiam, J., & Abbeel, P. (2020). Responsive safety in reinforcement learning by PID Lagrangian methods. In Proceedings of the 37th International Conference on Machine Learning (PMLR 119, pp. 9133–9143). <https://proceedings.mlr.press/v119/stooke20a.html>
- Wang, Z., Shi, X., Ma, C., Wu, L., & Wu, J. (2023). CCPO: Conservatively constrained policy optimization using state augmentation. In ECAI 2023 (Frontiers in Artificial Intelligence and Applications, Vol. 372, pp. 2599–2606). IOS Press. <https://doi.org/10.3233/FAIA230566>
- Welch, B. L. (1947). The generalization of "Student's" problem when several different population variances are involved. *Biometrika*, 34(1–2), 28–35. <https://doi.org/10.1093/biomet/34.1-2.28>
- Yao, Y., Liu, Z., Cen, Z., Zhu, J., Yu, W., Zhang, T., & Zhao, D. (2023). Constraint-conditioned policy optimization for versatile safe reinforcement learning. In Advances in Neural Information Processing Systems 36 (NeurIPS 2023). <https://arxiv.org/abs/2310.03718>

Appendix A. Settings Recorded at the Registration Commit

Settings verified against the OmniSafe documentation and on-policy benchmark before registration, and relied on in the body of this document.

Table A.1. Settings verified before registration.

Setting	Value	Where verified
Total steps; epochs; steps per epoch	10,000,000; 500; 20,000	OmniSafe on-policy benchmark
Actor and critic networks	Two hidden layers of 64 units; tanh	OmniSafe documentation, Lagrangian methods page
Update iterations; minibatch size; target KL (PPO-Lagrangian)	40; 64; 0.02	Published replication of OmniSafe defaults; confirmed at the commit
Cost limit; multiplier initial value; multiplier learning rate	25; 0.001; 0.035	OmniSafe documentation
Multiplier update	Once per epoch, equation (4)	OmniSafe PPOLag source
Surrogate objective	Equation (3)	OmniSafe PPOLag source
Device and parallelism	CPU; one torch thread; one vectorised environment; one process	OmniSafe on-policy benchmark
Episode length	1,000 steps	Ji et al. (2023)
Dormant threshold τ ; effective-rank δ	0.025; 0.01	Sokar et al. (2023); Kumar et al. (2021)

Copied verbatim from the repository at the registration commit, and not retyped here: the OmniSafe configuration file for PPO-Lagrangian on each task (learning rates, discount factors, GAE parameters, clipping, the multiplier optimiser, and every other field); the CPPOPID configuration file (PID gains, delays, moving-average factors, penalty maximum); the environment file with pinned versions of Safety-Gymnasium, OmniSafe, MuJoCo, PyTorch and Python; the workstation specification; the fixed evaluation batch for each task; and the PPO-Lagrangian reward on SafetyPointGoal1-v0 from the benchmark table, the one cell the extraction used for Table 3.5 did not render.

Appendix B. Results Ledger Schema

Table B.1. Fields of the results ledger.

Field	Content
run_id	Unique identifier
study; arm	A or B; task, N, onset shape, step-matching control, treatment, controller variant, or training-level set
seed	Integer from the fixed list
commit_hash; config_hash	Code and configuration at launch

Field	Content
started; finished; wall_clock_hours; machine	Timing and hardware
completed	True or false; if false, the cause (crash, incomplete, non-finite loss or multiplier)
checkpoints	Paths and steps
final_cost; final_return	Evaluation over 100 episodes at the final checkpoint
matched_checkpoint; training_age; selection_cost; measurement_cost; lambda_at_selection	Set by the comparison rule (Study A): the chosen checkpoint, its step index, its cost on the selection set and on the measurement set, and the multiplier at that point
gap_hazard; gap_dynamics; gap_finetune; gap_transfer	Robustness gaps of the matched checkpoint (Study A)
dormant_onset; rank_onset; norm_onset	Plasticity metrics at onset (Study A)
lambda_peak; lambda_final; settling_steps	Controller quantities (Study A)
sr_zero[budget]; sr_fewshot[budget, horizon]; adapt_steps[budget]	Satisfaction rates and adaptation steps (Study B)
notes	Free text; never used to exclude a run

Appendix C. Targeted Search Protocol for Study B

The search is registered here so that its result can be judged against a fixed procedure rather than a recollection. It is run before Study B's first run, its record is committed, and its path is entered in Table 9.0.

Table C.1. Search protocol.

Item	Specification
Question	Does any published or preprinted work vary the number or spacing of cost-budget levels seen during training as an independent variable and measure generalisation to unseen budgets?
Sources	Google Scholar, Semantic Scholar, arXiv, OpenReview, and the citation lists of Yao et al. (2023) and of every paper that cites it.
Queries	"constraint-conditioned" generalization threshold; "cost threshold" zero-shot safe reinforcement learning; "budget-conditioned" policy unseen; versatile safe reinforcement learning coverage; "cost limit" curriculum generalization unseen; each with and without "number of thresholds" and "spacing".
Inclusion	The training set of budgets is varied in number or spacing with the algorithm fixed, and an unseen budget is evaluated.
Exclusion	Works that vary the algorithm, that evaluate only seen budgets, or that vary several cost functions rather than levels of one.
Already known	Yao et al. (2023): three training budgets, unseen range, no variation of number or spacing. Sootla et al. (2022): single-budget training (60) against budgets sampled uniformly from 5 to 100, evaluated at 40 and 80; range-trained policies safer on unseen budgets; no variation of the number or spacing of discrete levels, no nearest-single-level comparison across several single levels, no adaptation cost. Wang et al. (2023, ECAI): several cost functions, not levels.
Record	For every hit: citation, what it varies, what it evaluates, and whether it meets the inclusion criterion; the date of the search; the searcher's name.
Decision	Any included work withdraws Study B by amendment; a partially overlapping work narrows Study B's claims by amendment, recorded before the first run.